Who benefits most?

Sorted partial effects · Journal Club / Open Discussion

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Our route

1. Choose the effect scale.
2. Explore the distribution and patient profiles.
3. Consider uncertainty and published evidence.
4. Discuss clinical decisions.

Choose the scale

01

A common odds ratio can hide very different absolute benefits.

↓ Start with two patients

Same treatment. Same benefit?

Both patient profiles have a **treatment odds ratio of 0.60**.

Lower baseline risk

10% → 6.25% event risk
Risk without → with treatment

Higher baseline risk

40% → 28.57% event risk
Risk without → with treatment

Which description of the treatment effect would you report?

Different absolute benefits

The same odds ratio. **Different absolute risk reductions.**

Lower baseline risk

3.75 pp
10% → 6.25% event risk

Higher baseline risk

11.43 pp
40% → 28.57% event risk

Heterogeneity depends on the effect scale we choose.

Define the benefit before sorting

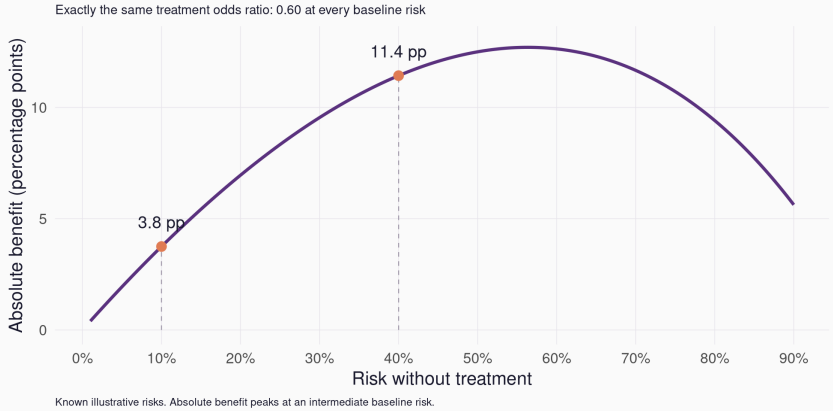
For a patient profile x:

$$\Delta(x) = \Pr(Y = 1 \mid A = 0, X = x) - \Pr(Y = 1 \mid A = 1, X = x)$$

- A binary **adverse outcome** at a fixed follow-up time.
- Larger values mean greater **absolute risk reduction**.
- A conditional average contrast for comparable profiles.

In our randomized-trial simulation, this identifies a conditional average causal benefit under the data-generating assumptions.

A constant odds ratio, varying absolute benefit



Simulation truth. The relationship can turn over at very high baseline risks; it is not globally increasing.

Look beyond the average

02

The same average benefit. Two very different distributions.

↓ Build the sorted-effects curve

One trial. Two effect patterns.

Scenario A

Constant treatment odds ratio of **0.60**.
Absolute benefit varies with baseline risk.

Scenario B

Treatment interacts with baseline severity.
Absolute benefit also reflects the interaction pattern.

The **true average absolute benefit is matched by design** in the same reference population.

Simulated randomized trial: 5,000 patients. Fixed reference population: 2,000 profiles. Logistic outcome models; no real patient data.

From a patient table to a sorted curve

Profile	Risk without treatment	Risk with treatment	Benefit
A	10%	6.25%	3.75 pp
B	20%	13.04%	6.96 pp
C	40%	28.57%	11.43 pp

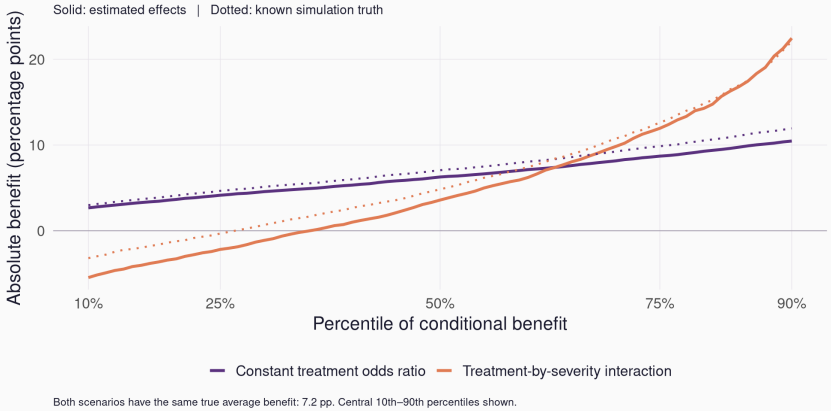
01 Predict twice

02 Subtract

03 Sort the contrasts

Here the contrast is a finite change in treatment, rather than a derivative. The same idea extends to other partial effects.

Similar averages can hide different distributions



True mean benefit: 7.2 pp in both scenarios. Negative benefit means increased adverse-event risk. Central 10th–90th percentiles shown.

What does a point on the curve mean?

The **80th percentile** is the benefit level at or below which approximately 80% of reference profiles lie.

It is a **quantile of conditional benefits**.

It is different from:

- the treatment effect on the 80th percentile of the outcome;
- a probability that a particular patient will benefit.

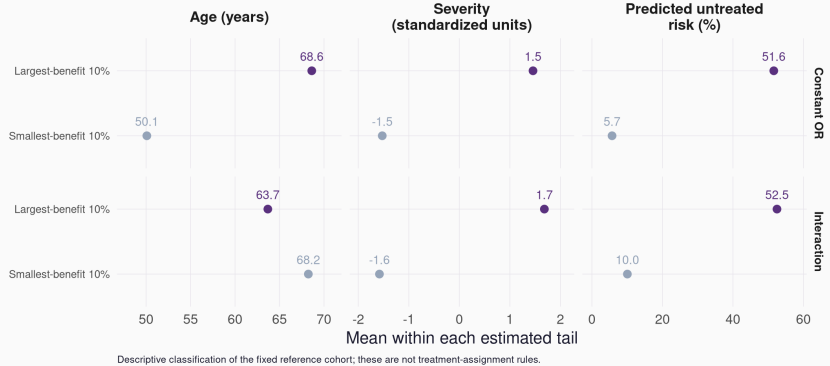
Meet the patient profiles

03

The curve shows how much. Classification asks who.

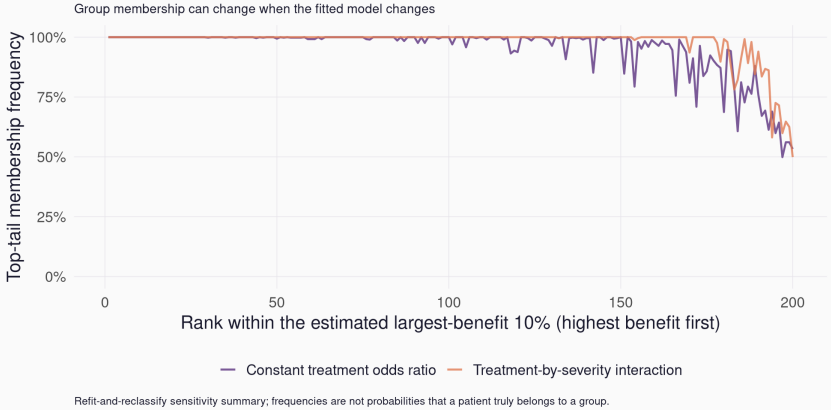
↓ Explore the tails

Who occupies the tails?



Descriptive profiles of the lowest and highest 10% of estimated benefits in both scenarios. These summaries have no classification confidence intervals.

A tail label can be unstable



Bootstrap reclassification in the fixed reference population. Membership frequencies measure sensitivity to refitting; they are not patient-level probabilities of benefit.

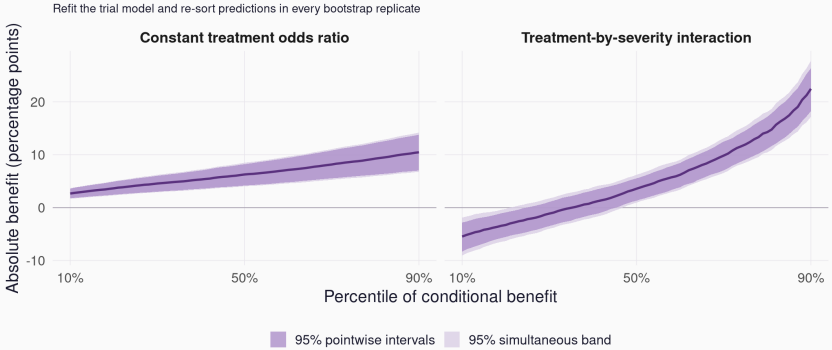
How stable is the picture?

04

A persuasive ranking still needs an uncertainty statement.

↓ Refit, re-sort, reconsider

Add uncertainty to the sorted curve



Approximate 95% intervals from refitting and re-sorting. Simultaneous coverage concerns the displayed percentile grid, conditional on the reference population and model.

What uncertainty have we included?

Included: estimation of the outcome model, prediction of contrasts, and re-sorting after each refit.

Held fixed: the reference patient profiles and the chosen model specification.

Requires further work: model sensitivity, new-population uncertainty, and formal inference for the clinical tail comparisons.

Would your main concern be sampling uncertainty or the model itself?

Read the evidence

05

What does the method reveal in a published application?

↓ A real-data example

A published example with real data

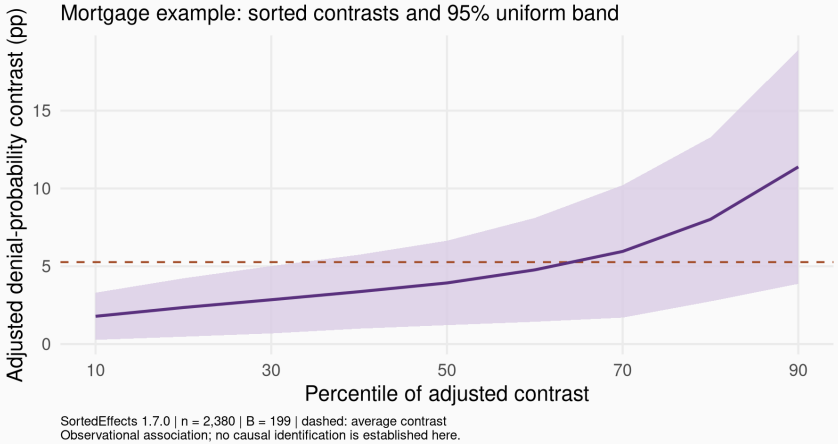
Mortgage denials in Boston, 1990 (Munnell et al. 1996; Chen et al. 2020).

- 2,380 applicants: 2,041 white and 339 Black applicants.
- Logistic model with financial and demographic controls.
- Contrast: predicted denial risk for Black versus white applicants at the same modeled covariates.

The tutorial reports an average adjusted denial disparity of **5.1 pp**, with a **90% interval of 2.1–8.1 pp**.

Published, bias-corrected result from Chen et al. (2020), Table 1; 500 bootstrap repetitions. This is a separate analysis from our clinical simulation.

From an average disparity to its distribution



New reproduction using the original SortedEffects 1.7.0 package. Positive values mean higher predicted denial risk for Black applicants. See validation notes for repetitions and comparisons.

From patterns to decisions

06

What additional evidence would justify changing treatment?

↓ Open the discussion

What would we do in a clinical study?

1. Specify the **outcome, time horizon, effect scale and population**.
2. Justify the **causal design** and fit a credible effect model.
3. Show the **average, sorted distribution and tail descriptions**.
4. Assess **uncertainty, model sensitivity and external validity**.

The sorted-effects method summarizes the effects supplied to it. Its interpretation inherits their assumptions.

Would you use this to choose treatment?

A patient profile lies in the highest-benefit 10%.

Before making a rule, consider:

- the **size** of benefit relative to harms and burden;
- uncertainty in the contrast and the **tail label**;
- whether the model transports to this setting;
- how the proposed rule performs in **independent evaluation**.

Three things to take away

Choose the scale.

Show the distribution.

Interrogate the tails.

An appealing ranking still needs a credible model, appropriate inference, and clinical validation.

Open discussion

1. Where would this add something to our current subgroup analyses?
2. Which assumptions or diagnostics would you want to challenge first?
3. What evidence would justify changing a treatment decision?

Keep the earlier plots available for discussion. Technical details and software checks follow in the backup slides.

Reading and resources

Method: (Chernozhukov, Fernández-Val, and Luo 2018).

Applied tutorial and original software: (Chen et al. 2020).

The session page contains the slides, code, numerical results and source notes.

Thank you — which clinical example should we try next?

A closer look

+

Optional details on the estimand, simulation and original software.

↓ Technical backup

Backup · What is being sorted?

$$\text{SPE}(u) = Q_{\Delta(X)}(u), \qquad \text{APE} = E[\Delta(X)].$$

- The distribution of X defines the population being summarized.
- Sorting discards the profile labels; classification reconnects them.
- A quantile of effects differs from a difference of outcome quantiles.

(Chernozhukov, Fernández-Val, and Luo 2018). Demo: empirical reference distribution; type-8 quantiles.

Backup · Simulation and fitting

Common untreated-risk model: baseline age, severity and sex.

A: treatment enters the logit with coefficient $\log(0.60)$.

B: treatment interacts with baseline severity; its intercept is calibrated to match A's true mean benefit.

Both fitted models use the corresponding correct specification.

R/demo.R contains the exact coefficients, random seeds, calibration, sample sizes and diagnostics.

Backup · Bootstrap ingredients

For each resample of the randomized trial:

1. Refit the outcome regression.
2. Predict both treatments for the **same** reference profiles.
3. Recalculate benefits and re-sort at each percentile.
4. Recalculate tail membership and descriptive summaries.

The simultaneous band uses the maximum standardized deviation over the displayed percentile grid.

Backup · Original software as a benchmark

- The original **SortedEffects 1.7.0** release is preserved in a pinned CRAN GitHub mirror.
- Reproduce its mortgage example using `spe()` and `ca()`.
- Compare point estimates with direct `glm()` predictions.
- Record bootstrap results, plotting behavior and limitations separately.

scripts/benchmark-sortedeffects.R and docs/software-validation.md. The clinical simulation and its inference target are separate from this benchmark.

References

Chen, Shuowen, Victor Chernozhukov, Iván Fernández-Val, and Ye Luo, "SortedEffects: Sorted causal effects in R," *The R Journal*, 12 (2020), 131–146.

Chernozhukov, Victor, Iván Fernández-Val, and Ye Luo, "The sorted effects method: Discovering heterogeneous effects beyond their averages," *Econometrica*, 86 (2018), 1911–1938.

Munnell, Alicia H., Geoffrey M. B. Tootell, Lynn E. Browne, and James McEneaney, "Mortgage lending in Boston: Interpreting HMDA data," *The American Economic Review*, 86 (1996), 25–53.